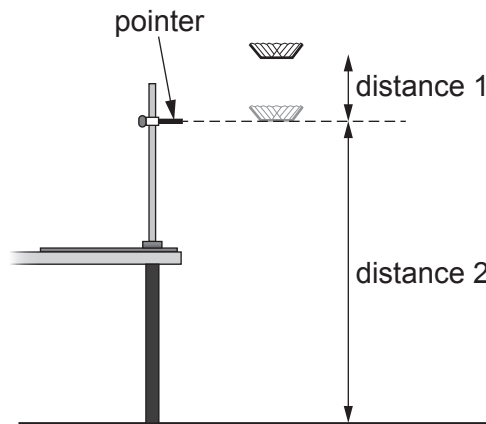


Examiner
only

4. A group of students is investigating how the terminal speed of cake cases is affected by the number being dropped.
They set up the apparatus as shown in the diagram below.



- (a) Describe a suitable method for this investigation.

[6 QER]

You should include in your answer:

- how the apparatus is set up
- suitable values for distances 1 and 2
- the measurements taken.



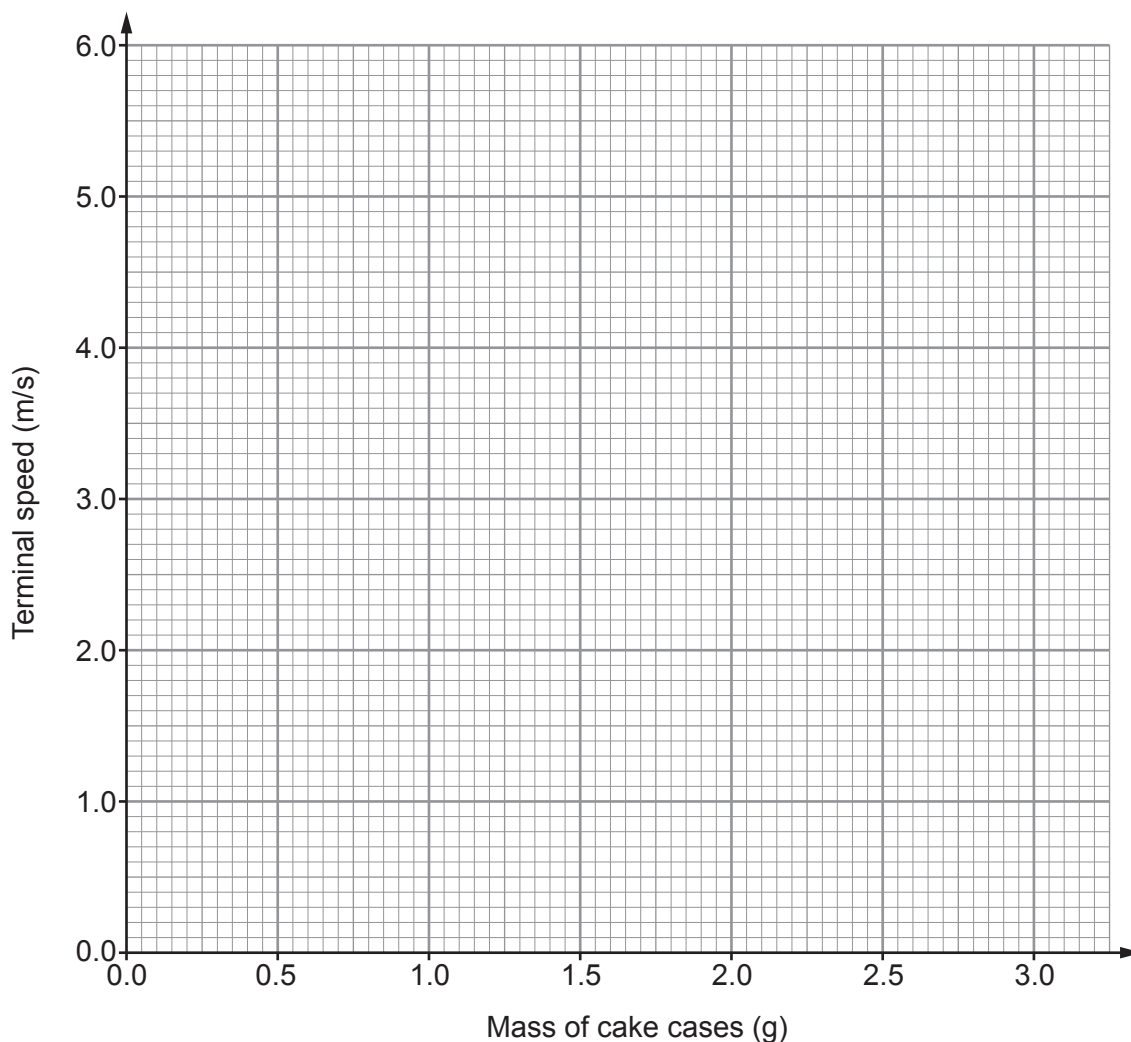
(b) The results from the investigation are shown in the table below.

Number of cake cases	Mass of cake cases (g)	Terminal speed (m/s)
0	0.0	0.0
2	1.0	2.5
3		4.8
4	2.0	4.3
5	2.5	5.0
6	3.0	5.5

(i) **Complete** the table. [1]



- (ii) I. **Plot** the data on the grid below. [2]
- II. **Circle** the anomalous result on the grid below. [1]
- III. **Draw** a suitable curve of best fit. [1]



- (iii) Use the graph to find the terminal speed for 1 cake case. [1]

terminal speed = m/s

